

Direct Next-Best-View Prediction from Frozen VGGT Geometry Features

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1. Introduction

Next-best-view (NBV) selection asks which camera view should be acquired next to reveal useful 3D information under a limited observation budget. This is relevant to active 3D reconstruction, where acquiring every possible view is expensive or impractical. The central computer-vision challenge is that the value of a candidate view is *history dependent*: a viewpoint may observe a large fraction of an object in isolation but add little once those surfaces have already been seen. A useful NBV policy must therefore infer object geometry from RGB observations, reason about occlusion and redundancy across views, and estimate which unseen camera will reveal new surface without access to ground-truth geometry at inference time.

PUN approaches this problem indirectly by predicting a Neural Uncertainty Map (NUM) from each acquired RGB image [9]. NUM supervision is obtained from single-view novel-view-synthesis error at 48 canonical cameras, and predictions from multiple observations are aggregated to select the next view, providing a task-specific uncertainty signal.

We investigate whether frozen VGGT representations [7] can provide a stronger basis for NBV selection. We separate three questions: First, can frozen representations predict the single-image NUM proxy? Second, does good proxy prediction actually transfer to sequential surface acquisition? Third, can NBV be learned more effectively by directly predicting the *marginal visible-surface gain* of each candidate from the complete observation history?

2. Method

Data and geometric target. We use NUM v2 RGB observations with paired ShapeNetCore v2 meshes [1, 9]. The object-disjoint split contains 870 training, 109 validation, and 300 test objects, each observed from the same 48 canonical camera anchors.

To obtain a reconstruction-backbone-independent NBV target, we precompute PUN-style Visible Area (VisA). For each object and camera v , a face-ID rasterization records the set $V(v)$ of directly visible mesh faces. For an acquired history H , area-weighted surface coverage is

$$C(H) = \frac{\sum_{f \in \cup_{v \in H} V(v)} A_f}{\sum_{f \in F} A_f},$$

where A_f is the area of face f . The marginal value of a candidate camera c is then

$$\Delta C(c | H) = C(H \cup c) - C(H).$$

Single-view proxy experiment. We first test whether frozen representations encode the original NUM signal. Frozen ImageNet ViT-B/16 [2], DINOv2 ViT-B/14 [6], and VGGT-1B features are mapped to the 48 NUM PSNR targets with lightweight prediction heads. We compare against an image-independent training-mean predictor and the released PUN checkpoint.

Sequential proxy transfer. We then deploy the validation-selected VGGT NUM predictor in closed loop. Every acquired image is processed independently, its source-relative PSNR map is aligned to the canonical cameras, and the maps are combined using the PUN-style filtering and product aggregation rule. This isolates whether successful prediction of the single-image NUM proxy is sufficient for sequential geometric exploration.

Direct history-conditioned NBV. Finally, we train frozen-VGGT policies directly on $\Delta C(c | H)$. The history dataset contains 34,800 training, 4,360 validation, and 12,000 test histories of lengths 1, 2, 4, 6, and 8. We evaluate four variants. *Independent* extracts each view with a separate frozen-VGGT forward and pools view features and camera directions. *Joint* processes the complete history jointly through VGGT before applying the same capacity-matched prediction head. These two models form the controlled comparison. *Pose DeepSets* preserves the association between each visual feature and its acquisition pose before pooling, while *Token attention* retains a 2×2 spatial token grid per view and scores each candidate through cross-attention. All locally trained heads use a masked Huber regression loss plus a pairwise ranking loss. VGGT remains frozen throughout.

Evaluation. Closed-loop evaluation uses the same 300 test objects, starts from camera 0, and allows 10 acquired views. We report final VisA coverage, area under the cover-

Table 1. Prediction performance for single-view NUM (top) and surface gain on 12,000 held-out histories (bottom). The blocks evaluate different targets and are not directly comparable.

Method	Regret ↓	Spearman ↑	NDCG@5 ↑
<i>Single-view NUM</i>			
Train mean	0.259	0.410	0.755
ImageNet-ViT	0.235	0.466	0.782
DINOv2	0.213	0.493	0.802
VGGT max-pool	0.211	0.509	0.807
PUN	0.169	0.602	0.846
<i>History-conditioned surface gain</i>			
Independent	0.176	0.707	0.849
Matched joint	0.199	0.661	0.827
Pose DeepSets [†]	0.162	0.720	0.861
Token Attn. [†]	0.151	0.744	0.873

age trajectory (AUC). Baselines include Random, a camera-space Farthest policy, and the released PUN checkpoint.

Because coverage does not necessarily imply good downstream reconstruction, we additionally reconstruct each selected view history with the same frozen VGGT model and a 2D Gaussian Splatting (2DGS) evaluation [3] and report normalized Chamfer-L1 against the ShapeNet surface.

3. Results

Frozen features encode useful single-view information. VGGT max-pooling achieves the strongest NUM ranking among locally trained frozen representations. (Table 1). Its advantage over DINOv2 is modest, however, and ImageNet-ViT achieves lower Huber error. The task-specific PUN checkpoint remains stronger on all tabulated metrics. Thus, frozen VGGT features encode useful NBV-related information but are neither uniformly superior to other frozen representations nor sufficient to match PUN on its original proxy task.

Proxy accuracy does not ensure sequential exploration. When deployed through PUN-style alignment and aggregation, the VGGT NUM predictor gives the lowest coverage and AUC: 0.816 and 5.821 (Table 2). Final coverage falls below Random (0.851), Farthest (0.862), and PUN (0.850). This separates single-view feature decodability from sequential decision quality. NUM estimates reconstruction difficulty from an individual observation, whereas NBV requires additional visible surface conditioned on the full history.

Direct surface-gain supervision substantially improves NBV. All four history-conditioned policies outperform the VGGT proxy in coverage and AUC, and exceed Random, Farthest, and PUN in AUC. These results support directly learning history-conditioned geometric gain rather

Table 2. Ten-view coverage and reconstruction. 2DGS CD uses a smaller subset.

Method	VisA ↑	AUC ↑	VGGT	2DGS
			CD ↓	CD ↓
Random	0.851	6.184	0.218	0.0367
Farthest	0.862	6.233	0.220	0.0287
PUN	0.850	6.380	0.322	0.0428
VGGT proxy	0.816	5.821	0.349	0.0334
Oracle	0.885	6.993	0.218	0.0436
Independent	0.864	6.783	0.199	0.0263
Matched joint	0.862	6.754	0.211	0.0327
Pose DeepSets [†]	0.865	6.816	0.175	0.0217
Token Attn. [†]	0.867	6.832	0.223	0.0238

than transferring independently predicted NUMs.

Joint backbone processing alone does not help. Independent and matched joint use capacity-matched heads, providing the controlled architectural comparison. Joint processing slightly lowers Huber error but worsens on the other metrics. Coverage and AUC also decline. Moreover, joint selection requires reprocessing the full history through VGGT rather than reusing independently cached features. We therefore find no advantage from joint processing in this frozen-backbone, pooled-head configuration. The larger diagnostic heads perform better, which suggests benefits from retaining feature–pose associations or spatial tokens. However, their greater trainable capacity prevents attributing the improvements solely to these architectural choices.

Coverage and reconstruction quality have different rankings. Pose DeepSets produces the lowest Chamfer, despite slightly lower coverage than token attention. The true-gain Oracle achieves the highest coverage yet does not minimize either reconstruction error. VisA is therefore useful as a reconstruction-backbone-independent exploration metric, but maximizing visible surface does not necessarily yield the best reconstruction. The 2DGS results additionally require caution because they cover only a test subset.

Limitations and implications. The larger diagnostic heads are not capacity-matched controls, and small differences among history-conditioned variants should be interpreted cautiously. VGGT reconstruction also depends on camera-based similarity alignment, with occasional alignment failures. Finally, evaluation uses synthetic ShapeNet/NUM imagery; real-world transfer remains untested.

Overall, frozen VGGT features support effective NBV when trained to predict history-dependent surface gain. Single-view proxy accuracy alone does not ensure a strong sequential policy, and joint frozen-backbone processing alone offers no observed benefit. Preserving pose and spatial information is promising, but coverage and downstream reconstruction must remain separate evaluation criteria.

A. Related Work

Next-best-view selection. Classical NBV methods typically maintain an explicit geometric state and evaluate candidate viewpoints. NBV-Net [5] instead learns to predict a discrete next view from a 3D occupancy representation, reducing online search but still requiring an explicit voxelized state. PUN [9] predicts a 48-anchor uncertainty map directly from each RGB observation and combines predictions across acquired views. Our Phase-2 setting follows this 48-view formulation, but asks whether a frozen geometry-aware representation can replace a task-specific uncertainty encoder. Phase 3 further differs by predicting true history-conditioned surface gain rather than a single-image uncertainty proxy.

Frozen visual and geometric representations. Vision Transformer (ViT) [2] and DINOv2 [6] provide strong generic image representations, whereas VGGT [7] is explicitly trained for feed-forward multi-view geometry. This motivates our comparison between generic frozen features, independently encoded VGGT observations, and joint multi-view VGGT processing. AREA3D [8] also uses feed-forward 3D perception for active reconstruction, but constructs an explicit 3D uncertainty field and combines geometric confidence with vision-language guidance. In contrast, we directly decode candidate-view utility from frozen VGGT features without building an explicit learned 3D uncertainty field. We evaluate geometric coverage using ShapeNet meshes [1], and separately test whether improved visibility translates to improved reconstruction quality.

Gaussian-splatting reconstruction. Gaussian splatting provides a complementary reconstruction endpoint to the frozen-VGGT point clouds. 3D Gaussian Splatting represents scenes with optimized volumetric Gaussian primitives for novel-view synthesis [4], whereas 2D Gaussian Splatting constrains the primitives to oriented surface elements for improved geometric accuracy [3]. We therefore use 2DGS for quantitative geometry evaluation and retain 3DGS for qualitative appearance comparison.

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